University Course Management System

Phase 0

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1)Scope Statement:

The University Course Management System is a software application designed for a large academic institution that needed a single organized platform to manage course-related activities efficiently. Before this system, handling enrollments, assignments, grades, and user access without a dedicated system was difficult and inefficient. The University Course Management System was built to solve this by bringing all course-related operations into one place.

The system supports four types of users. Administrators manage user permissions, course categories, and data consistency. Faculty Members create and manage course content, set assignments, and monitor student progress. Registered Students can enroll in courses, submit assignments, and access their grades, while Guest Users can only view general course information without enrolling.

The system handles role-based access, assignment submissions with file uploads and deadlines, grade tracking, and automatic notifications to faculty when submissions are ready. All user data, course information, and grades are stored securely. The system is built to support up to 5,000 concurrent users, applies data encryption for sensitive information, and performs regular backups to prevent any data loss. The system will not cover payment processing, live video streaming, physical classroom scheduling, or university admissions as these fall outside the boundaries of the University Course Management System.

2) Functional Requirements:

Administrator:

- The system shall allow the administrator to manage user permissions.
- The system shall allow the administrator to create course categories .
- The system shall allow the administrator to update course categories.
- The system shall provide the administrator with tools to monitor, detect, and resolve data inconsistencies across all system records.

Faculty Member:

- The system shall allow the faculty member to create, edit and delete course content.
- The system shall allow the faculty member to create assignments including a title, description, file submission requirements and a submission deadline.
- The system shall allow the faculty member to monitor student progress by viewing assignment submission statuses, grades, and overall course completion rates per student.
- The system shall generate notifications for faculty member when assignment submissions are ready for grading.

Registered student:

- The system shall allow the registered student to browse and enroll in available courses.
- The system shall allow the registered student to view course content within the enrolled courses.
- The system shall allow the registered student to submit assignments by uploading files before the specified deadline.
- The system shall allow the registered student to view their grades and faculty feedback for all graded assignments within their enrolled courses.

Guest User:

- The system allow the guest user to view general course information including course title, description, and category without requiring registration or login.

3) Non-Functional Requirements:

Performance:

- The system must support 5000 concurrent users without performance degradation

Security:

- All sensitive data should be encrypted
- Each user should only access the data permitted for them, 100% of unauthorized access attempts must be blocked

Reliability:

- The System uptime must be 99%

Usability:

- Students must be able to enroll in course in less than 2 minutes
- Faculty members and students must be able to upload assignments in less than 2 minutes

4) Project Boundaries:

What the system will do

1. The system will allow students to browse and enroll in available courses.
2. The system will enable faculty members to create, manage, and grade assignments.
3. The system will store and manage user data, course content, and grades securely.

What the system will not do

1. The system will not handle financial transactions or course payments.
2. The system will not provide live lectures or real-time video streaming.
3. The system will not manage physical classroom scheduling or on-campus resources.